The Chain-of-Thought Tax Is Paid in Tokens, Not Accuracy: A Within-Family Scale Study of When Reasoning Traces Help

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Abstract

Chain-of-thought (CoT) prompting is used by default, yet it costs far more output tokens than a direct answer. We test a concrete version of a common belief: that zero-shot CoT lowers accuracy on pattern matching and factual recall by 5–15 points, raises it on logical deduction and arithmetic by 10–25 points, and helps fewer task categories as models grow. We run a paired comparison of direct answering and CoT on five sizes of one model family (QWEN2.5-INSTRUCT, 0.5B to 14B parameters) over 8 tasks and 2,350 items per model, with item-level bootstrap intervals and Holm-corrected McNemar tests. The hypothesis mostly fails. On pattern matching and factual recall the CoT effect lies between -2.2 and $+3.8$ points and is never significantly negative. Arithmetic gains 48 to 85 points against a strict no-working baseline, and logical deduction ranges from -6.9 points at 0.5B to $+28.7$ points at 14B. The scale trend runs opposite to the hypothesis: the number of tasks that CoT significantly helps rises from 2 of 8 to 4 of 8, and the share of previously correct answers that CoT breaks falls with size in every category. In this setting the tax is the roughly $75\times$ output-token cost for no measurable gain on classification; the only accuracy loss appears for the smallest model on multi-step logic, partly because its reasoning runs past the token budget. Our results cover one model family up to 14B parameters and do not speak to frontier-scale models.

1 Introduction

Asking a language model to "think step by step" is now a default prompting habit. The habit has a visible price: a reasoning trace takes far more output tokens than a bare answer. If the trace also lowered accuracy on some kinds of task, practitioners would be paying twice, and routing prompts by task type would be worth real engineering effort.

There is reason to suspect such a cost. Chain-of-thought (CoT) gains concentrate on mathematical and symbolic problems Wei et al. [2022], Kojima et al. [2022], Sprague et al. [2024], and CoT can hurt on tasks where deliberation also hurts people, such as implicit pattern learning Liu et al. [2024]. A natural extrapolation is that CoT taxes tasks that depend on recognition or recall, and that the tax grows with model size because larger models do more of the work internally.

Does zero-shot CoT actually cost accuracy on easy task types, and does its benefit shrink with scale? We test a specific, falsifiable form of this claim, split into three parts. Let $\Delta = \mathrm{acc}(\mathrm{CoT}) - \mathrm{acc}(\mathrm{DIRECT})$, in accuracy points. **H1**: $\Delta \in [-15, -5]$ for pattern matching and factual recall. **H2**: $\Delta \in [+10, +25]$ for logical deduction and arithmetic. **H3**: fewer categories benefit as parameter count grows.

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Existing evidence does not settle these questions for two reasons. First, most comparisons mix model families, so scale is confounded with training data and recipe. Second, an accuracy difference near zero does not distinguish "CoT changes nothing" from "CoT breaks some answers and repairs as many others".

We address both. We compare direct answering and CoT on five sizes of a single family, QWEN2.5-INSTRUCT from 0.5B to 14B parameters, on eight tasks in four categories (2,350 items per model). Every item is answered under both conditions, so we report paired statistics, and alongside the accuracy difference we report a *harm rate*, $P(\text{CoT wrong} \mid \text{direct right})$, and a *fix rate*, $P(\text{CoT right} \mid \text{direct wrong})$.

The hypothesis mostly fails. On pattern matching and factual recall, Δ ranges from -2.2 to $+3.8$ points across ten model-by-category cells; none is significantly negative and none falls in the hypothesised range. CoT helps arithmetic by 48 to 85 points, far above the hypothesised range, because answering arithmetic with no visible working is near 0% accurate. On logical deduction Δ climbs from -6.9 points at 0.5B to $+28.7$ points at 14B. Only 1 of 20 cells is both inside its hypothesised range and significant. The scale trend is the reverse of H3: CoT significantly helps 2 of 8 tasks at 0.5B and 4 of 8 at 14B, and harm rates fall with size in every category. What CoT does cost is tokens: 71–87$\times$ more output for no measurable gain on classification.

In summary, our contributions are:

- We run a within-family scale ladder (five sizes, a roughly 30$\times$ parameter range) that separates the effect of scale on CoT from differences between model families.
- We show that zero-shot CoT carries no measurable accuracy tax on text classification or factual recall in this family, and bound any such tax below 5 points for nine of ten cells.
- We show that the benefit of CoT grows with scale on logical deduction ($+7.6$ points per doubling of parameters), and that harm rates fall with scale in all four categories.
- We trace the one negative cell (the 0.5B model on logical deduction) in part to runaway reasoning that exhausts the token budget, and we document the limits of our evidence, including run-to-run churn and an incomplete robustness check.

We review related work in section 2, describe the setup in section 3, present results in section 4, and discuss interpretation and limitations in section 5.

2 Related Work

We did not run a systematic literature search for this study; this section positions our work against the papers that motivated it.

Chain-of-thought prompting. Wei et al. [2022] show that few-shot exemplars containing intermediate reasoning steps improve performance on arithmetic, commonsense and symbolic tasks, and Kojima et al. [2022] show that a single zero-shot instruction to think step by step yields similar gains. We use the zero-shot form, because it is what practitioners apply by default and it needs no task-specific exemplars.

Where CoT helps. Sprague et al. [2024] conduct a large meta-analysis and new experiments and find that CoT gains concentrate on math and symbolic reasoning, with little benefit elsewhere. Their finding for other tasks is mainly an absence of gain. Building on this, we ask the sharper question of whether CoT is *harmful* on recognition and recall tasks, and we use paired item-level outcomes to separate a null effect from offsetting harms and fixes.

Where CoT hurts. Liu et al. [2024] show that CoT can reduce performance on tasks where verbal deliberation makes humans worse, such as implicit pattern learning. Unlike their task selection, our "pattern matching" category consists of standard text classification, which models already solve near ceiling. Our null result on classification therefore does not contradict theirs; it shows that the harm they find does not extend to ordinary classification in the models we test.

Scale. Most comparisons of CoT across model sizes span different families, so scale is confounded with training data and recipe. We hold the family fixed and vary only size, using the five smallest instruction-tuned sizes of Qwen2.5 Qwen Team [2024].

Table 1: Prompt conditions. The suffix follows the item text and its answer-format hint.

Condition	Instruction suffix	Prefill	Max tokens
DIRECT	Give only the final answer, with no explanation or reasoning. Respond in the format "Answer: <answer>".	"Answer:"	24
COT	Think step by step and write out your reasoning. Then finish with a final line in the format "Answer: <answer>".	none	1024
COT-V2	Before answering, carefully work through the problem and explain your thought process. When you are done, end your response with one last line of the form "Answer: <answer>".	none	1024

Table 2: Tasks. Chance is the accuracy of uniform guessing.

Category	Task	Answer format	Chance
Pattern matching	AG NEWS topic Zhang et al. [2015]	4 labels	25%
	SST-2 sentiment Socher et al. [2013]	2 labels	50%
Factual recall	TRIVIAQA, no context Joshi et al. [2017]	free-form, alias match	≈0%
	MMLU-RECALL, 7 subjects Hendrycks et al. [2021]	4-way multiple choice	25%
Logical deduction	BBH-LOGIC, five objects Suzgun et al. [2023]	5-way multiple choice	20%
	PROOFWRITER, depth 5 Tafjord et al. [2021]	True / False / Unknown	33%
Arithmetic	GSM8K Cobbe et al. [2021]	number	≈0%
	SYNTHARITH (ours)	integer	≈0%

3 Methodology

Problem setup. For a model m, a task t and an item i, let $y^{\text{dir}}_{m,t,i}, y^{\text{cot}}_{m,t,i} \in \{0,1\}$ be correctness under the direct and CoT conditions. The CoT effect on a task is the paired difference $\Delta_{m,t} = \frac{1}{n_t}\sum_i (y^{\text{cot}}_{m,t,i} - y^{\text{dir}}_{m,t,i})$, reported in accuracy points. The category-level effect is the unweighted mean of the two task-level effects in that category. We read the hypothesised "5–15%" and "10–25%" as accuracy points.

Models. We use QWEN2.5-INSTRUCT at 0.5B, 1.5B, 3B, 7B and 14B parameters Qwen Team [2024], run locally with bfloat16 weights and greedy decoding, zero-shot, with the system prompt "You are a helpful assistant." We originally planned to use API models; no API key was available, so we ran open-weight models instead. This removes frontier models from the study but gives a cleaner scale comparison: one family and one training recipe.

Prompt conditions. Each item ends with an answer-format hint (e.g., "The answer must be exactly one of: World, Sports, Business, Sci/Tech."). A condition adds one instruction suffix (table 1). In DIRECT we prefill the assistant turn with "Answer:" and cap generation at 24 tokens, so the model cannot write any reasoning; we choose this strict baseline because it gives the cleanest contrast with CoT. COT-V2 is a paraphrase used only for a robustness check on the first 100 items of each task. Greedy decoding has no sampling seed, so prompt wording is the nuisance variable we check.

Tasks. We use two tasks for each of four categories (table 2), sampling 300 items per task with a fixed seed; the BBH task has 250 items and we use all of them, for 2,350 items in total. MMLU-RECALL pools seven recall-heavy MMLU subjects (miscellaneous, world religions, high school geography, prehistory, anatomy, nutrition and astronomy). SYNTHARITH consists of three-operation integer expressions such as $(125 + 859) \times 4 - 93$ that we generate for this study, so they cannot appear verbatim in a training set. Assigning tasks to categories is a judgement call; we return to this in section 5.1.

Scoring. We take the answer to be the text after the last "Answer:" or "answer is" marker (the first line of output in DIRECT). Classification outputs must match exactly one allowed label, multiple-choice outputs must give the right letter, numbers are compared numerically, and TRIVIAQA uses standard normalisation with alias containment. A CoT response with no parsable answer counts as wrong. We audited extraction by reading samples of wrong-scored CoT answers for each task. The audit found one artefact, which we fixed before computing the reported numbers: the 0.5B model sometimes wrote "Science/Tech" for the label "Sci/Tech", in both conditions.

Table 3: CoT effect Δ by category and model size, in accuracy points, with 95% bootstrap intervals. *marks Holm-significant cells ($\alpha = 0.05$, 20 tests). The one cell that is both inside its hypothesised range and significant is in **bold**.

Category	0.5B	1.5B	3B	7B	14B	Hypothesised
Pattern matching	-0.7 [$-3.2, +1.8$]	-1.2 [$-3.2, +0.8$]	$+0.7$ [$-1.2, +2.5$]	-0.2 [$-1.5, +1.2$]	0.0 [$-1.2, +1.2$]	-15 to -5
Factual recall	-2.2 [$-5.8, +1.5$]	-0.2 [$-3.5, +3.2$]	$+1.8$ [$-1.2, +4.8$]	$+3.8$ [$+1.2, +6.5$]	$+2.8$ [$+0.3, +5.3$]	-15 to -5
Logical deduction	-6.9 [$-12.3, -1.5$]	$+7.2$ [$+2.2, +12.3$]	**$+13.9$*** [$+8.8, +18.8$]	$+26.7$* [$+21.8, +31.6$]	$+28.7$* [$+24.2, +33.3$]	$+10$ to $+25$
Arithmetic	$+47.7$* [$+43.7, +51.7$]	$+77.0$* [$+73.7, +80.3$]	$+84.7$* [$+81.8, +87.3$]	$+85.0$* [$+82.2, +87.7$]	$+81.7$* [$+78.8, +84.5$]	$+10$ to $+25$

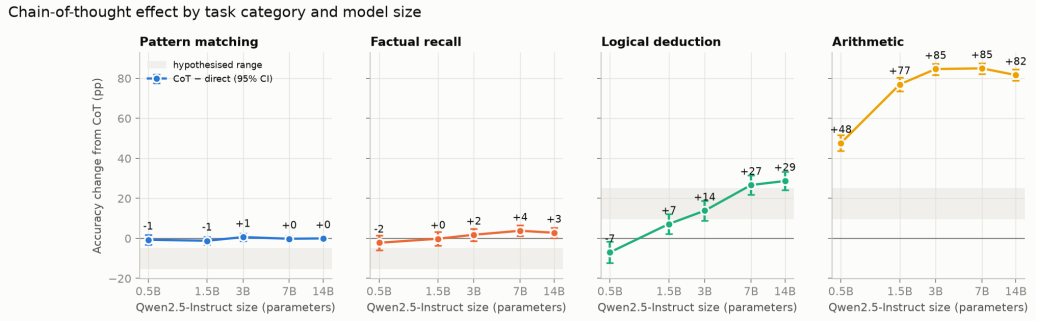


Figure 1: CoT effect (CoT minus direct accuracy, in points) by category and model size, with 95% bootstrap intervals. Grey bands mark the hypothesised ranges. Pattern matching and factual recall stay near zero and never enter their band; logical deduction rises through its band with size; arithmetic sits far above its band.

Metrics. Besides accuracy and Δ, we report the harm rate $P(y^{\text{cot}} = 0 \mid y^{\text{dir}} = 1)$ and the fix rate $P(y^{\text{cot}} = 1 \mid y^{\text{dir}} = 0)$, the share of CoT responses that lack an answer marker or hit the token budget, and mean output tokens.

Statistics. We compute 95% bootstrap confidence intervals for Δ over items (10,000 resamples, stratified by task) and exact McNemar tests McNemar [1947], with Holm correction Holm [1979] applied separately to the 40 task-level tests and the 20 category-level tests at $\alpha = 0.05$. We count an H1 or H2 cell as supported if its point estimate lies in the hypothesised range and the cell is Holm-significant. For H3 we fit the slope of the category-level Δ on $\log_2$(parameters), with a confidence interval from the item bootstrap, and we count the tasks with a significant benefit at each size.

Deviations from the pre-run plan. The analysis plan was written before the runs. Two things differ from it. The plan set the CoT budget at 640 tokens; the runs used 1,024. The plan judged H1 and H2 by whether the bootstrap interval excluded zero; we report the stricter Holm-corrected criterion. The second change does not alter any verdict: under either criterion the same single cell is supported.

4 Results

4.1 No accuracy tax on pattern matching or factual recall (H1)

Table 3 and figure 1 give the category-level effect of CoT. For pattern matching, Δ lies between -1.2 and $+0.7$ points at every size, and the lower confidence bound is never below -3.2, so the main prompt excludes a 5-point tax. For factual recall, Δ lies between -2.2 and $+3.8$ points. A 5-point tax is excluded from 1.5B up; at 0.5B the interval $[-5.8, +1.5]$ just reaches it. None of the ten cells is in the hypothesised range and none is significantly negative. At 7B and 14B, recall moves slightly in favour of CoT ($+3.8$ and $+2.8$); the bootstrap intervals exclude zero but neither cell survives Holm correction. TRIVIAQA drives this movement (table 4).

The closest thing to H1 at the task level is the 0.5B model on MMLU-RECALL (-6.0 points, interval $[-13.0, +1.0]$, not significant). No task shows a Holm-significant CoT penalty at any size.

Table 4: Task-level accuracy (%): direct → CoT, with Δ in parentheses. Holm-significant Δ ($\alpha = 0.05$, 40 tests) in **bold**. No task shows a significant CoT penalty at any size.

Task	0.5B	1.5B	3B	7B	14B
AG NEWS	83.0 → 84.3 (+1.3)	83.3 → 79.7 (−3.7)	81.3 → 83.3 (+2.0)	88.3 → 87.3 (−1.0)	86.7 → 87.3 (+0.7)
SST-2	87.7 → 85.0 (−2.7)	92.0 → 93.3 (+1.3)	93.3 → 92.7 (−0.7)	93.7 → 94.3 (+0.7)	95.7 → 95.0 (−0.7)
TRIVIAQA	18.7 → 20.3 (+1.7)	40.0 → 42.7 (+2.7)	47.3 → 48.7 (+1.3)	55.0 → 62.0 (+7.0)	66.0 → 70.0 (+4.0)
MMLU-RECALL	48.7 → 42.7 (−6.0)	69.7 → 66.7 (−3.0)	75.0 → 77.3 (+2.3)	83.7 → 84.3 (+0.7)	86.3 → 88.0 (+1.7)
BBH-LOGIC	24.0 → 18.8 (−5.2)	26.4 → 33.2 (+6.8)	43.6 → 50.0 (+6.4)	54.4 → 74.8 (**+20.4**)	64.4 → 86.8 (**+22.4**)
PROOFWRITER	33.3 → 24.7 (−8.7)	35.7 → 43.3 (+7.7)	29.7 → 51.0 (**+21.3**)	31.3 → 64.3 (**+33.0**)	39.7 → 74.7 (**+35.0**)
GSM8K	4.3 → 37.3 (**+33.0**)	7.0 → 77.7 (**+70.7**)	14.0 → 89.0 (**+75.0**)	21.7 → 94.7 (**+73.0**)	32.0 → 97.3 (**+65.3**)
SYNTHARITH	0.0 → 62.3 (**+62.3**)	0.0 → 83.3 (**+83.3**)	0.0 → 94.3 (**+94.3**)	0.0 → 97.0 (**+97.0**)	0.3 → 98.3 (**+98.0**)

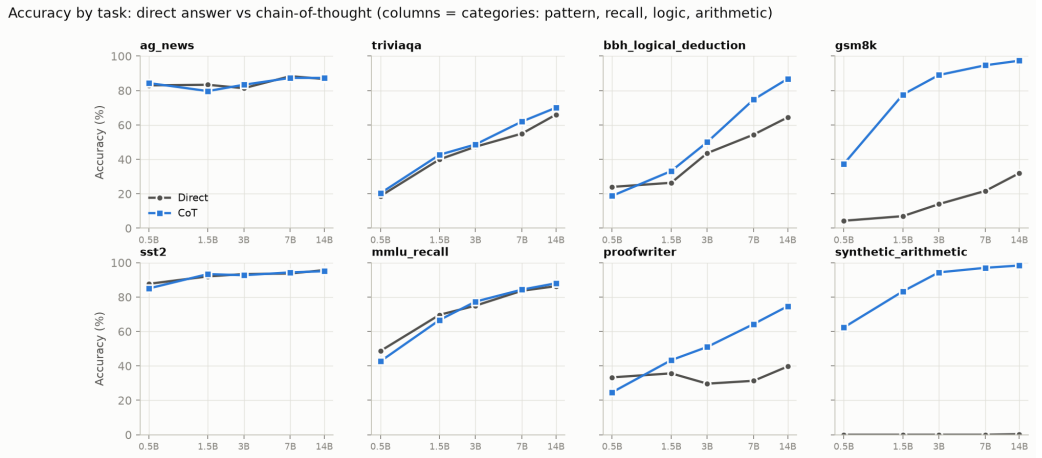


Figure 2: Accuracy by task for direct answering and CoT. Columns are categories (pattern matching, factual recall, logical deduction, arithmetic). The two conditions overlap on classification and recall; they separate with size on logic; on arithmetic the direct condition is near zero throughout.

4.2 Large gains on arithmetic, size-dependent gains on logic (H2)

The direction of H2 holds from 1.5B up, but the magnitudes do not match. Arithmetic gains are 2 to 8 times larger than the hypothesised 10–25 points. This follows from the strict baseline: with the answer prefilled, models cannot show any working, and DIRECT accuracy on SYNTHARITH is 0.0–0.3% at every size (table 4, figure 2). Logical deduction is in range only at 3B (+13.9). It is negative at 0.5B, below the range at 1.5B, and above it at 7B and 14B.

PROOFWRITER shows the contrast most clearly. DIRECT accuracy stays near chance (29.7–39.7%) at every size, while CoT accuracy climbs from 24.7% to 74.7%.

Four task-level cells have uncorrected $p < 0.05$ but do not survive correction: PROOFWRITER at 0.5B (−8.7, $p = 0.049$), AG NEWS at 1.5B (−3.7, $p = 0.027$), PROOFWRITER at 1.5B (+7.7, $p = 0.038$) and TRIVIAQA at 7B (+7.0, $p = 0.0025$, Holm $p = 0.061$).

4.3 The benefit of CoT grows with scale (H3)

H3 predicts that fewer categories benefit as models grow. We find the opposite (table 5). CoT significantly helps 2 of 8 tasks at 0.5B and 1.5B, 3 at 3B, and 4 at 7B and 14B. No slope of Δ on $\log_2$(parameters) is negative. The slope is +7.6 points per doubling for logical deduction and +1.2 for factual recall, and it is indistinguishable from zero for pattern matching.

The arithmetic slope (+6.7) needs a caveat: it comes almost entirely from the step from 0.5B to 1.5B. From 3B up the arithmetic Δ is flat to slightly falling (+84.7, +85.0, +81.7). On GSM8K, Δ peaks at 3B (+75.0) and falls to +65.3 at 14B, because DIRECT accuracy keeps rising (14.0% to 32.0%) while CoT accuracy approaches its ceiling (89.0% to 97.3%).

Table 5: Scale trend. *(Left)* Number of tasks and categories with a Holm-significant CoT effect at each size. *(Right)* Slope of Δ per doubling of parameters, with 95% bootstrap intervals.

	0.5B	1.5B	3B	7B	14B
Tasks helped (of 8)	2	2	3	4	4
Tasks hurt (of 8)	0	0	0	0	0
Categories helped (of 4)	1	1	2	2	2

Category	Slope
Pattern matching	$+0.2$ $[-0.3, +0.7]$
Factual recall	$+1.2$ $[+0.3, +2.0]$
Logical deduction	$+7.6$ $[+6.3, +8.9]$
Arithmetic	$+6.7$ $[+5.7, +7.6]$

Table 6: Harm rate $P(\text{CoT wrong} \mid \text{direct right})$ and fix rate $P(\text{CoT right} \mid \text{direct wrong})$, in %. Arithmetic harm rates rest on at most 8 harmed items per cell.

	Harm rate					Fix rate				
Category	0.5B	1.5B	3B	7B	14B	0.5B	1.5B	3B	7B	14B
Pattern matching	6.1	4.4	2.7	1.6	1.3	30.7	21.6	23.7	14.8	13.2
Factual recall	36.1	17.0	10.6	5.5	5.0	15.1	20.3	21.5	25.0	28.0
Logical deduction	88.8	45.1	35.9	18.7	9.6	26.4	31.3	42.9	60.3	69.6
Arithmetic	61.5	9.5	0.0	3.1	1.0	50.1	80.1	91.0	95.7	97.6

4.4 Harm and fix rates

A Δ near zero can hide offsetting changes, so we look at the two directions separately (table 6, figure 3). Harm rates fall with size in every category: from 6.1% to 1.3% for pattern matching, 36.1% to 5.0% for factual recall, and 88.8% to 9.6% for logical deduction. Fix rates rise with size in every category except pattern matching, where they fall from 30.7% to 13.2%.

Two cautions apply. The arithmetic harm rates rest on very few items, because few arithmetic items are right in the direct condition (8 harmed items at 0.5B and 0–2 at other sizes). And part of every harm and fix rate is run-to-run churn rather than an effect of CoT (section 5.1), so Δ is the safer number.

4.5 Output diagnostics and token cost

Token cost. Mean output length is 3.1–3.3 tokens per item for DIRECT and 222–271 for CoT, a ratio of 71$\times$ to 87$\times$ depending on the model.

Well-formedness. CoT responses carry an explicit answer marker and finish within budget in at least 95% of cases for every model-task pair, with two exceptions on the smallest models. On SYNTHARITH the 0.5B model omits the marker in 24% of responses; it ends with a boxed number, which the fallback parser reads. On PROOFWRITER the small models often run out of budget: 44.0% of CoT responses hit the 1,024-token limit at 0.5B, 10.7% at 1.5B, 5.0% at 3B, 0.3% at 7B and none at 14B.

Where does the 0.5B loss on logic come from? The 0.5B model has 142 logical-deduction items that are right under DIRECT and wrong under CoT. Of these, 48 are truncated runs in which the model restates rule after rule until the budget runs out, 1 has no answer marker, and 93 give a wrong final answer. When we restrict to items whose CoT response is well-formed, logical deduction at 0.5B moves from -6.9 to $+3.7$ points ($n = 407$ of 550, interval $[-3.8, +11.1]$). All other cells move by less than 2 points. This subset is selected on the outcome of the CoT run, so it shows where the loss comes from; it is not an unbiased estimate of what CoT would do with a larger budget.

4.6 Prompt robustness

We repeat the comparison with the paraphrased instruction COT-V2 on the first 100 items of each task (table 7 in the appendix). Across the 19 available model-by-category cells, the two wordings' effects correlate at 0.996. Signs agree in 16 of 19 cells, and in the three disagreements both effects are within 4 points of zero.

This check has three weaknesses that we state plainly. First, it is incomplete for 14B: out-of-memory errors on shared GPUs repeatedly killed that run, which finished 544 of 800 items and no logical-

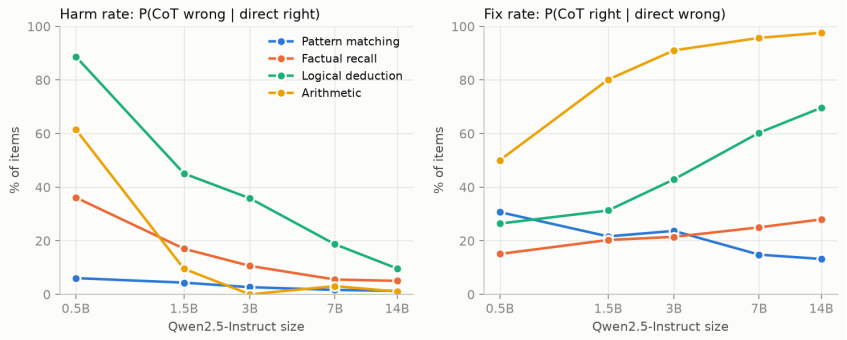


Figure 3: Harm rates *(Left)* and fix rates *(Right)* by category and model size. Harm rates fall with size in all four categories. Fix rates rise with size except for pattern matching.

deduction items. Second, with CoT-v2, pattern matching shows small negative effects whose uncorrected intervals exclude zero for three models (0.5B: -2.5 $[-5.0, -0.5]$; 1.5B: -2.5 $[-5.0, -0.5]$; 14B: -2.3 $[-4.7, -0.5]$) and a positive one for 3B ($+3.5$ $[+0.5, +7.0]$). These are uncorrected, rest on 200 or fewer items, and are mixed in sign, so we read them as bounding any classification penalty at roughly 2–5 points and not as evidence for one. Third, on this 100-item subset the 0.5B effect on logical deduction is positive under both wordings ($+3.0$ and $+3.5$, with a CoT-v2 interval of $[-5.0, +12.0]$), unlike the -6.9 on the full set. The negative cell in table 3 is therefore not stable across subsets of items.

5 Discussion

What is the chain-of-thought tax? In this family, the cost of CoT on recognition and recall tasks is tokens, not accuracy. For classification, CoT spends roughly 75× more output tokens for no measurable change. For recall it buys at most a few points at the larger sizes. Routing prompts by task type is therefore justified here on cost grounds, but not as protection against a loss of accuracy.

Why does H2 miss on magnitude? The arithmetic gap depends on how strict the baseline is. We prefill the answer, so the model can do no visible working at all, and three-operation arithmetic then fails almost every time. A looser "be concise" baseline without prefill would likely shrink the gap; we did not test it.

Do larger models "internalise" reasoning that traces disrupt? The original hypothesis ends with this mechanism claim, which accuracy data cannot test directly. One pattern resembles it: on GSM8K the CoT gain shrinks from $+75.0$ at 3B to $+65.3$ at 14B as direct accuracy rises. Larger models do more of the arithmetic without a trace. But this is a shrinking benefit under a ceiling, not a trace disrupting anything; CoT is still 65 points ahead at 14B. Everything else points the other way: harm rates fall with size, and larger models make better use of an explicit trace on logic. Whether the gain keeps shrinking, or turns negative, in much larger models is untested here.

What goes wrong when CoT hurts? The only category-level cell with a negative interval is logical deduction at 0.5B, and it is not significant after correction (Holm $p = 0.15$). About a third of the harmed items are runaway outputs that hit the token budget (section 4.5). For the rest, the 0.5B model's direct accuracy on these tasks is at chance (24.0% and 33.3%), so most of its "correct" direct answers are likely lucky guesses, and losing them is weak evidence of disrupted reasoning. Our reading is that a very small model cannot carry out the reasoning, and attempting it costs a few points through runaway outputs.

Individual harms on easy tasks do occur. Asked for the term for a tribe with no set homeland, the 0.5B model reasoned from the obvious "Nomadic" to "Schematic". The 14B model talked itself from "Sci/Tech" to "Business" for an article about Microsoft entering online music. At 14B such cases are balanced or outnumbered by fixes: 7 harmed and 7 fixed items for pattern matching, 23 harmed and 40 fixed for recall.

Two observations we did not expect. Direct accuracy on PROOFWRITER barely moves with scale (33.3% to 39.7%); the 14B model answers "Unknown" for about two thirds of items when it cannot reason visibly. And the fix rate on pattern matching falls with size (30.7% to 13.2%) while it rises elsewhere. With direct accuracy near 91% at 14B, we suspect the remaining errors are mostly ambiguous labels that reasoning cannot resolve, but we have not checked this.

5.1 Limitations

One family, at most 14B parameters. The hypothesis concerns scale in general. Our results cover Qwen2.5 from 0.5B to 14B only. We did not test frontier models, and the claim that very large models are disrupted by traces remains open.

Not bit-reproducible at the item level. Batched bfloat16 generation depends slightly on batch composition. Re-running 128 items per model in different batches gave identical CoT text for only about 25% of items; item-level correctness agreed for 85% (0.5B) and 95% (14B) of CoT items and 99% of direct items. Our bootstrap intervals resample items only, so they understate total uncertainty somewhat, and harm and fix rates include this churn. We did not run repeated full passes to quantify it.

Strict direct baseline. Prefilling "Answer:" removes all visible reasoning. This gives the cleanest contrast but inflates arithmetic gains relative to common prompting practice.

Category assignment. "Pattern matching" here means text classification, which these models perform at 83–96%. We did not include tasks built around implicit pattern learning, where prior work reports real harm from CoT Liu et al. [2024]; they may behave differently. The ceiling on SST-2 limits how much CoT could gain, though not how much it could lose.

Token budget. The 1,024-token limit truncated 44% of the 0.5B model's PROOFWRITER responses; a larger budget might change that cell.

Sample size and robustness. With 250–300 items per task, task-level intervals span roughly ± 3 to ± 8 points, and we cannot resolve category effects smaller than about 3 points. The prompt-robustness check uses 100 items per task, is incomplete for 14B, and covers only one paraphrase.

Other scope limits. All public benchmarks may appear in training data, which affects both conditions; SYNTHARITH is the only contamination-free task. We test zero-shot prompting, greedy decoding and English only. Finally, our account of prior work rests on four papers we knew in advance and not on a systematic review.

6 Conclusion

We tested whether zero-shot chain-of-thought prompting taxes accuracy on easy task types and helps less as models grow, using five sizes of QWEN2.5-INSTRUCT on eight tasks. It does neither in this family. CoT did not reduce accuracy on text classification or factual recall (effects between -2.2 and $+3.8$ points, none significantly negative), improved arithmetic by 48–85 points against a no-working baseline and logical deduction by up to 29 points, and helped more tasks, not fewer, as size grew. The tax we measure is the roughly $75\times$ output-token cost for no gain on classification, plus an accuracy loss for the smallest model on multi-step logic that is not significant after correction and is tied in part to truncated outputs.

The most valuable next step is to repeat the study on frontier models and a second open family, to test whether the shrinking GSM8K benefit becomes a penalty at larger scale. Further steps are to add tasks built around implicit pattern learning, to run repeated passes so that harm and fix rates carry proper error bars, to add a concise no-prefill baseline, and to raise the token budget for the smallest models to separate "cannot reason" from "ran out of room".

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A Prompt Robustness Details

Table 7 lists the effect of each CoT wording on the robustness subset (first 100 items of each task). The 14B rows cover fewer items because the CoT-v2 run for that model did not finish (section 4.6).

B Per-Task Heatmap

C Compute and Implementation

We ran all models on four NVIDIA RTX A6000 GPUs (48 GB each) that were shared with other jobs, using Hugging Face `transformers` and PyTorch with bfloat16 weights. Batch sizes were 64–96 and were halved automatically on out-of-memory errors. The main grid took about 1 h 50 min of wall-clock time under contention; the 14B CoT pass alone took about 55 min. We sampled items with seed 42. The analysis re-scores every response from the saved raw outputs.

Table 7: CoT effect on the robustness subset for the main wording and the paraphrase, in accuracy points, with the 95% bootstrap interval for the paraphrase (uncorrected). The 14B model has no logical-deduction items for CoT-v2.

Model	Category	n	CoT	CoT-v2	CoT-v2 95% CI
0.5B	Pattern matching	200	-4.0	-2.5	$[-5.0, -0.5]$
	Factual recall	200	-0.5	-0.5	$[-6.0, +5.0]$
	Logical deduction	200	$+3.0$	$+3.5$	$[-5.0, +12.0]$
	Arithmetic	200	$+50.0$	$+58.0$	$[+51.0, +65.0]$
1.5B	Pattern matching	200	-1.5	-2.5	$[-5.0, -0.5]$
	Factual recall	200	$+4.0$	-2.0	$[-7.5, +3.5]$
	Logical deduction	200	$+9.0$	$+5.5$	$[-3.0, +14.0]$
	Arithmetic	200	$+76.0$	$+75.5$	$[+69.5, +81.0]$
3B	Pattern matching	200	$+1.5$	$+3.5$	$[+0.5, +7.0]$
	Factual recall	200	$+4.0$	$+3.5$	$[-2.5, +9.5]$
	Logical deduction	200	$+15.0$	$+16.5$	$[+8.0, +25.0]$
	Arithmetic	200	$+82.5$	$+79.5$	$[+74.0, +85.0]$
7B	Pattern matching	200	-1.0	0.0	$[-2.0, +2.0]$
	Factual recall	200	$+4.5$	$+3.0$	$[-1.5, +7.5]$
	Logical deduction	200	$+27.0$	$+24.0$	$[+15.0, +33.0]$
	Arithmetic	200	$+84.5$	$+83.0$	$[+77.5, +88.0]$
14B	Pattern matching	178	-1.0	-2.3	$[-4.7, -0.5]$
	Factual recall	175	$+3.2$	-0.2	$[-4.0, +3.5]$
	Arithmetic	191	$+80.3$	$+76.6$	$[+70.7, +82.2]$

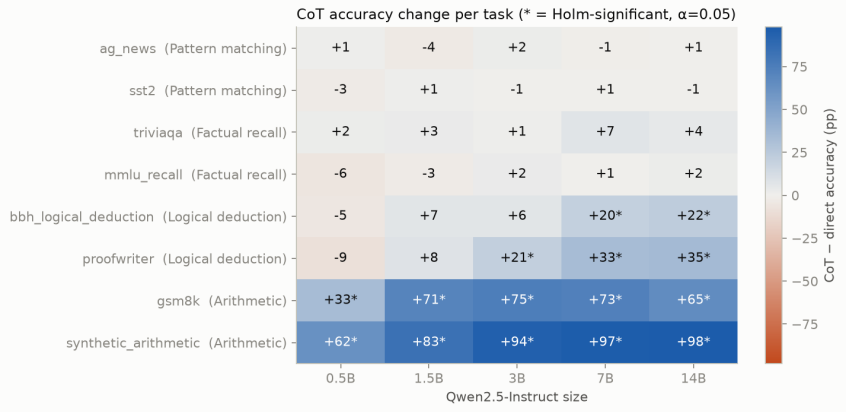


Figure 4: CoT effect for every task and model size, in accuracy points. This is the same information as table 4.